

Data Cleaning, Analysis and Visualization with Excel 2026

1. Project Overview

This project presents an end-to-end analysis of a comprehensive retail dataset. By integrating customer demographics, product inventory, and store location data with transactional records, this study provides a holistic view of the business's performance across various segments. The goal is to uncover patterns in consumer behavior and operational efficiency to support data-driven decision-making.

2. Project Objectives

- **Revenue Analysis:** Evaluate total revenue performance across different regions and store types.
- **Product Performance:** Identify top-performing product categories and sub-categories.
- **Customer Segmentation:** Analyze purchasing behavior based on loyalty levels (Platinum, Gold, Silver, Bronze) and demographic factors (Age, Gender).
- **Regional Strategy:** Compare regional sales to identify growth opportunities and underperforming areas.

3. Dataset Description

The dataset is structured into four primary relational tables and a summary pivot table:

- **Sales:** Contains 2,000 transactional records including Order_Date, Quantity, Unit_Price, Discount, and Total_Amount.
- **Customer:** Detailed profiles of customers including Name, Age, Gender, City, State, and Loyalty_Level.
- **Product:** Catalog information including Product_Name, Category (Electronics, Clothing, Sports, etc.), Sub_Category, and Cost.
- **Store:** Operational details for 20 stores, categorized by Region (East, West, North, South) and Store_Type (Flagship, Outlet, Online).
- **Pivot Table:** Aggregated summaries of revenue by region and customer spending patterns.

4. Data Preparation & Transformation

- **Data Cleaning:** Handled missing values and ensured consistency in naming conventions across the datasets (e.g., standardizing Customer_ID and Product_ID).

- **Data Integration:** Merged the four primary tables into a master analytical dataset using key identifiers to link transactions with specific customers, products, and stores.
- **Aggregation:** Calculated total transaction values, applied discounts, and summarized data into pivot formats for regional and categorical analysis.

5. Tools and Technologies

- **Excel:** Used for initial data inspection and generating pivot-based summaries.
- **Data Visualization:** Conceptualized using standard analytical plotting to visualize trends.

6. Analysis and Insights

- **Financial Overview:** Total revenue generated is approximately 2,175,107 across 2,000 transactions, with an average order value of 1,087.55.
- **Category Performance:** **Sports** (541,932) and **Electronics** (504,311) are the leading categories, significantly outperforming Beauty and Clothing.
- **Regional Performance:** The **North** (647,266) and **South** (635,213) regions are the primary revenue drivers. The **West** region shows significant potential for growth as it currently lags behind others.
- **Loyalty Insights:** **Platinum** members contribute the highest share of revenue (646,364), followed by Bronze and Gold members.
- **Demographics:** Female customers account for a higher total spending (1.20M) compared to Male customers (0.94M).

7. Key Outcomes

- **High-Value Segments:** Platinum loyalty members and the female demographic represent the most profitable customer segments.
- **Product Strength:** The business has a strong foothold in the Sports and Electronics markets.
- **Regional Concentration:** Sales are heavily concentrated in the North and South, suggesting a need for targeted marketing in the West region to balance the portfolio.

8. Conclusion

The analysis reveals a robust retail operation with clear strengths in specific product categories and regional markets. The high contribution from Platinum loyalty members suggests that the current retention programs are effective. Moving forward, the business should focus on expanding its presence in the West region and leveraging the high average order value in the Sports and Electronics categories to further increase profitability.